

FMCG Supply Chain Analytics

Executive Dashboard & Operational Performance Report

Domain: FMCG / Supply Chain **Primary Tool:** Tableau Desktop **Scope:** Sales, Procurement, Inventory & Risk

TOTAL SALES REVENUE

\$33.43M

Across All Regions

PROCUREMENT COST

\$21.47M

Total Purchase Outlay

INVENTORY VALUE

\$525.24M

Cumulative Carrying Value

REORDER RISK

5.52%

Overall Exposure Rate

1. Executive Summary

This project focuses on analyzing Fast-Moving Consumer Goods (FMCG) supply-chain data to provide an executive-level overview of sales, procurement expenditure, inventory carrying value, demand forecasting accuracy, and warehouse replenishment risk. An interactive dashboard was developed using **Tableau** to consolidate critical operational KPIs and analytical views into a single, cohesive interface.

The dashboard empowers supply-chain stakeholders to move beyond fragmented, siloed data. It enables proactive monitoring of operational performance, immediate identification of potential stockout risks, evaluation of forecast precision against actual sales, and deep-dive analysis of monthly inventory and procurement patterns.

2. Business Problem & Objectives

FMCG supply chains operate in high-velocity environments, generating vast datasets across multiple domains—including sales transactions, purchase orders, warehouse stock levels, SKUs, and forecast models. Analyzing these streams in isolation creates severe operational visibility gaps.

Key Operational Challenges

- Difficulty in verifying if actual demand aligns with forecast planning models.
- Uncertainty surrounding monthly inventory fluctuations and capital tie-up.
- Lack of prioritized risk rankings for regional distribution centers.
- Disconnected tracking of procurement expenditure versus inventory demand.

Core Project Objectives

- Track aggregate sales revenue and regional performance distribution.
- Monitor procurement expenditure trends across monthly planning cycles.
- Evaluate cumulative inventory value against stock movement over time.
- Isolate high-risk warehouses and assess top-volume SKUs against reorder thresholds.

3. Tools & Analytical Technologies

Tableau Desktop was leveraged as the core business intelligence and dashboard development platform. The project utilized advanced analytical features, including custom calculated fields, dynamic parameters, dynamic sorting, synchronized multi-axis trendlines, and customized layout containers.

4. KPI Summary & Interpretation

KPI Name	Result	Business Context & Interpretation
Total Sales Revenue	\$33.43M	Reflects overall gross revenue generated across all product categories and geographic zones represented in the dataset.
Total Procurement Cost	\$21.47M	Quantifies total purchasing expenditure committed to raw materials and stock replenishment during the analyzed period.
Total / Cumulative Inventory Value	\$525.24M	Represents the aggregate financial value of stock held across warehouses. Provides a benchmark for working capital evaluation.
Overall Reorder Risk	5.52%	Standardized risk metric tracking the proportion of inventory items/events approaching critical safety stock thresholds.

5. Detailed Dashboard Visual Components

5.1 Actual vs Forecast Demand Purpose: Compares actual units sold against forecasted demand across monthly planning windows. Finding: Actual demand tracks forecast demand with remarkable accuracy. This validates the reliability of existing demand planning models.	5.2 Monthly Inventory Value Trend Purpose: Tracks cumulative inventory valuation shifts across the calendar year. Finding: Highlights significant capital accumulation mid-year, followed by stabilization, indicating clear seasonal inventory build cycles.
5.3 Monthly Procurement Cost Trend Purpose: Tracks purchasing expenditure cycles month-by-month. Finding: Identifies specific cost spikes that can be evaluated against demand peaks to ensure purchasing synchronicity.	5.4 Reorder Risk by Warehouse Purpose: Ranks distribution facilities based on calculated reorder vulnerability. Finding: WH_5 exhibits the highest relative reorder risk among all facilities, pinpointing an immediate operational priority.
5.5 Sales Revenue by Region Purpose: Evaluates geographic revenue contributions. Finding: Highlights key revenue-generating territories, supporting optimized regional safety stock allocation.	5.6 Inventory Level vs Reorder Point Purpose: Compares stock levels against reorder triggers for the Top 10 SKUs. Finding: Concentrating on top-tier SKUs provides executive clarity while flagging specific high-value items requiring replenishment.

6. Key Business Findings

- **Strong Forecast Convergence:** Predictive demand planning models are operating effectively, providing a solid baseline for procurement scheduling.
- **Localized Facility Exposure:** Warehouse **WH_5** represents the primary operational bottleneck, demonstrating significantly higher reorder exposure than other fulfillment nodes.
- **Capital Efficiency Opportunities:** With cumulative inventory valuation at \$525.24M relative to \$33.43M in sales revenue, stock velocity and inventory turnover warrant detailed SKU-level audits.

Critical Operational Focus: Warehouse WH_5

While WH_5 shows the highest reorder risk exposure (5.52% global rate context), this does not automatically indicate active stockouts. However, it mandates immediate review of local safety stock thresholds, vendor lead times, and fulfillment cycle efficiency.

7. Strategic Business Recommendations

1. Prioritize Target Monitoring at WH_5

Initiate a deep-dive operational audit into WH_5 replenishment lead times, vendor compliance rates, and local safety stock parameters to mitigate elevated risk profiles.

2. Leverage Forecast Models for Working Capital Optimization

Since actual sales closely track forecasts, leverage predictive models to negotiate synchronized, dynamic procurement schedules with suppliers, avoiding excess capital tie-up.

3. Conduct ABC/XYZ Inventory Segmentation

Categorize the \$525.24M inventory pool using ABC/XYZ analysis to separate essential safety buffers from slow-moving or excess inventory items.

4. Automate Exception Alerts for Top SKUs

Establish automated dashboard notifications when Top 10 SKU stock levels approach within 10% of their established reorder boundaries.

8. Repository & Deliverables Architecture

The recommended project organization for deployment and version control:

```
FMCG-Supply-Chain-Analytics/
├── README.md # Executive Summary & Setup Guide
├── Tableau/
│   └── FMCG_Supply_Chain_Executive_Dashboard.twbx # Editable Packaged Workbook
├── Dashboard/
│   └── FMCG_Supply_Chain_Dashboard.png # High-Resolution Dashboard Capture
├── Data/
│   └── processed_supply_chain_dataset.csv # Cleaned Analytical Dataset
├── Documentation/
│   └── FMCG_Supply_Chain_Project_Report.pdf # Official PDF Project Report
```

9. Conclusion

This project demonstrates the powerful application of supply chain analytics and data visualization in Fast-Moving Consumer Goods. By integrating sales, procurement expenditure, inventory movement, forecasting accuracy, warehouse risk, and SKU reorder dynamics into a single Tableau interface, the business gains complete operational transparency. The resulting dashboard transforms raw operational data into actionable strategic intelligence.